

Ans to the Q. No - 2 (a)

We know,

$$T(n) = 2T\left(\frac{n}{2}\right) + n \quad \text{--- (1)}$$

Let,

$$n = 2^m$$

$$\begin{aligned} \therefore T(2^m) &= 2T\left(\frac{2^m}{2}\right) + 2^m \\ &= 2T(2^{m-1}) + 2^m \quad \text{--- (ii)} \end{aligned}$$

Now,

$$T(2^{m-1}) = 2T(2^{m-2}) + 2^{m-1} \quad \text{--- (iii)}$$

~~putting~~ putting the value in eq (ii) $\rightarrow$

$$\begin{aligned} T(2^m) &= 2[2T(2^{m-2}) + 2^{m-1}] + 2^m \\ &= 4T(2^{m-2}) + 2^m + 2^m \quad \text{--- (4)} \end{aligned}$$

Now,

$$T(2^{m-2}) = 2T(2^{m-3}) + 2^{m-2} \quad \text{--- (5)}$$

putting the value in eq (4) $\rightarrow$

$$\begin{aligned} T(2^m) &= 4[2T(2^{m-3}) + 2^{m-2}] + 2^m + 2^m \\ &= 8T(2^{m-3}) + 2^m + 2^m + 2^m \end{aligned}$$

☐☐

$$\Rightarrow T(2^m) = 2^m T(2^{m-m}) + m \cdot 2^m$$

$$= 2^m + m \cdot 2^m$$

$$~~= 2^m (1 + m)~~$$

$$= 2^m (1 + m)$$

Now,

$$n = 2^m$$

$$\Rightarrow m = \lg_2 n$$

$$\therefore T(n) = n(1 + \lg_2 n)$$

$$\Rightarrow T(n) = n + n \lg_2 n$$

$$\Rightarrow T(n) = n \lg_2 n$$

(Ans.)